

Vadodara Institute of Engineering

Diploma Engineering–Semester –3–Mid-Examination –Summer-2026

Subject Code: DI04016051

Subject Name: MAD

Time: 1:00 PM TO 03:30 PM

Date: 28-04-2026

Branch: IT

Total Marks: 70

Instructions:

1. Attempt all questions.
2. Make Suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Use of programmable & Communication aids are strictly prohibited.
5. Use of non-programmable scientific calculator is permitted.
6. English version is authentic.

		Marks
Q.1	(a) What is Android?	03
	(અ) Android શું છે?	03
	(b) Explain features of Android OS	04
	(બ) Android ની features સમજાવો.	04
	(c) Compare Android and iOS.	07
	(ક) Android અને iOS નો તફાવત સમજાવો.	07
	OR	
	(c) Explain Android versions and their features	07
	(ક) Android ની versions અને તેમની features સમજાવો	07
Q.2	(a) What is Android Studio?	03
	(અ) Android Studio શું છે?	03
	(b) Explain Android architecture	04
	(બ) Android architecture સમજાવો	04
	(c) Explain all layers of Android architecture in detail	07
	(ક) Android architecture ની બધી layers વિગતવાર સમજાવો.	07
	OR	
Q.2	(a) What is Emulator?	03
	(અ) Emulator શું છે?	03
	(b) Explain Application Framework in Android.	04
	(બ) Application Framework સમજાવો.	04
	(c) Explain Android Runtime and Linux Kernel.	07
	(ક) Android Runtime અને Linux Kernel સમજાવો	07
Q.3	(a) What is Activity?	03
	(અ) Activity શું છે?	03
	(b) Explain Activity lifecycle.	04
	(બ) Activity lifecycle સમજાવો.	04
	(c) Explain all application components (Activity, Service, Broadcast Receiver, Content Provider)	07
	(ક) બધી application components સમજાવો.	07
	OR	
Q.3	(a) What is Service?	03

	(અ) Service શું છે?	03
	(b) Explain Service lifecycle	04
	(બ) Service lifecycle સમજાવો	04
	(c) Explain Android Manifest file with its role	07
	(ક) Android Manifest અને તેનો ઉપયોગ સમજાવો	09
Q. 4	(a) What is Layout?	03
	(અ) Layout શું છે?	03
	(b) Explain different layouts (Linear, Relative, Constraint).	04
	(બ) વિવિધ layouts સમજાવો.	04
	(c) Explain Views and Widgets with examples.	07
	(ક) Views અને Widgets ઉદાહરણ સાથે સમજાવો	09
OR		
Q. 4	(a) What is Event-driven programming?	03
	(અ) Event-driven programming શું છે?	
	(b) Explain event listeners and callback methods.	04
	(બ) Event listeners અને callback methods સમજાવો.	04
	(c) Explain event handling methods (onClick, onKey, onTouch, onFocusChange)	07
	(ક) Event handling methods સમજાવો.	09
Q.5	(a) What is Intent?	03
	(અ) Intent શું છે?	03
	(b) Explain types of Intents (Explicit and Implicit).	04
	(બ) Intents ની પ્રકાર સમજાવો.	04
	(c) Explain data storage methods (SharedPreferences, SQLite, Firebase).	07
	(ક) Data storage methods (SharedPreferences, SQLite, Firebase) સમજાવો.	09
OR		
Q.5	(a) What is Firebase?	03
	(અ) Firebase શું છે?	03
	(b) Explain SQLite database and CRUD operations.	04
	(બ) SQLite અને CRUD operations સમજાવો. 04	04
	(c) Explain recent trends (Kotlin, Flutter, React Native, APIs, JSON).	07
	(ક) Recent trends (Kotlin, Flutter, React Native, APIs, JSON) સમજાવો.	09